UNIVERSITY SCHOLARSHIP MANAGEMENT SYSTEM

Software Requirements Specification

(Prepared in accordance with IEEE Std 830-1998)

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Signature Page

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Change History

Version	Date	Author	Changes Made
1.0	7 Oct,2026	Laiba Iftikhar	Initial draft of the SRS
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Remarks:

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of the University Scholarship Management System. The system will provide a centralized platform for managing scholarship information, eligibility criteria, advertisements, student applications, application status, and interviews. This SRS provides a clear and structured description of the system requirements and serves as a reference for the development, testing, and evaluation of the system.

Purpose of the system:

To develop a centralized system for managing university scholarships for specific programs, including scholarship information, advertisements, student applications, and approval status.

1.2 Scope

Project name: University Scholarship Management System

1.2.1 Project Summary

The University seeks to develop a University Scholarship Management System to maintain and manage information about scholarships offered to students. The university offers different scholarships under two basic categories: Need-Based and Merit-Based. Each scholarship is associated with a specific academic program, such as BS. The scholarship section will be responsible for advertising available scholarships and managing student applications. Students will be able to view and apply for eligible scholarships through the system. Once a student's scholarship application is approved, the student will not be allowed to receive or apply for any other scholarship. The system will provide a centralized platform for managing scholarship information, advertisements, applications, and approval status.

1.2.2 Scope of the System

- Student Authentication: Authenticate students according to their academic program.
- Scholarship Categorization: Categorize scholarships into Need-Based and Merit-Based Scholarships according to programs.
- Eligibility Criteria: Define and check the eligibility criteria for each scholarship.
- Program Based Eligibility: Allow students to submit applications for scholarships specific to their academic program.
- Interview Scheduling: Schedule interviews for eligible scholarship applicants when required.
- Application Approval: Manage the approval or rejection of scholarship applications.
- Scholarship Restriction: Ensure that a student cannot receive another scholarship once their application has been approved.
- Scholarship Information Importing: Import or receive scholarship information and updates from external scholarship portals, such as the HEC Portal and PEEF Portal.
- Scholarship Advertisement: Advertise available scholarships to inform students about scholarship opportunities and their application deadlines.
- Application Status: Allow students to view the status of their scholarship applications, including whether an application is pending, approved, or rejected.

1.2.3 Objectives

- Categorize scholarships according to the student's academic program and scholarship type.
- Ensure scholarship eligibility by applying defined eligibility criteria.
- Prevent multiple scholarships by keeping track of students who have already received a scholarship.
- Advertise scholarships on time so that eligible and deserving students do not miss opportunities.
- Manage scholarship applications and keep track of their application status.
- Schedule interviews for eligible scholarship applicants.
- Manage scholarship approvals and maintain accurate records of approved applications.

1.2.4 Out of Scope

The following functions will not be handled by our system:

- Paying scholarship money to students.
- Managing admissions, fees or exam results because our system is only concerned with scholarship management.
- Scholarships offered by outside organizations that are not registered in the scholarship management section.

1.3 Definitions, Acronyms, and Abbreviations

1.3.1 Definitions

#	Term	Definition
1	Scholarship	Financial assistance provided to a student to support their educational expenses.
2	Need-Based Scholarship	A scholarship awarded to students based on their financial need.
3	Merit-Based Scholarship	A scholarship awarded to students based on their academic performance or achievements.
4	Eligibility Criteria	The requirements that a student must meet to qualify for a particular scholarship.
5	Scholarship Application	A formal request submitted by a student to apply for a scholarship.
6	Scholarship Advertisement	Information published by the university to inform students about available scholarships.
7	Application Status	The current state of a scholarship application, such as pending, approved, or rejected.

#	Term	Definition
8	Scholarship Approval	The process of accepting a student's application and granting the scholarship.
9	Authorized Staff	University staff members who are permitted to manage scholarship information and applications.
10	Interview Scheduling	The process of arranging interviews for students who are eligible for a scholarship.
11	Gantt Chart	A visual representation of the project schedule, showing tasks, durations, and dependencies.

1.3.2 Acronyms and Abbreviations

Acronym	Meaning
SRS	Software Requirements Specification
UAT	User Acceptance Testing
IEEE	Institute of Electrical and Electronics Engineers
UI	User Interface
DBMS	Database Management System
CGPA	Cumulative Grade Point Average
UC	Use Case
BS	Bachelor of Science

1.4 References

- ISO/IEC/IEEE 16326: Systems and Software Engineering – Project Management
- Software Engineering: A Practitioner's Approach – Roger S. Pressman
- HEC Scholarship Management System (e-Portal) — for scholarship applications and management.

1.5 Overview

The remaining part of this document is organized as follows. Section 2 gives an overall description of the product, including its perspective, functions, user characteristics, constraints, and assumptions. Section 3 contains the specific requirements, including external interfaces, functional requirements, performance requirements, database requirements, design constraints, and software system attributes. The appendices contain supporting material such as use case diagrams and the domain models.

2. Overall Description

2.1 Product Perspective

The University Scholarship Management System is a centralised web-based system for the university's scholarship section. It is a centralised platform for handling scholarship notices, application forms and approval status. It is linked to the university's administration and system so that students remain updated with newly announced scholarships.

Basically, our system connects following three concerned users:

- Students, who log in to view scholarships, apply and check their status.
- Authorized scholarship staff, who add scholarships, review applications, schedule interviews and approve or reject applications.
- The university administration system, which provides student information and announced scholarships.

The scholarship system will be linked with the university administration to receive and display newly announced scholarships.

2.1.1 System Interfaces

The system receives student information (name, registration number, academic program) and newly announced scholarships from the university administration system. No other external system is needed.

2.1.2 User Interfaces

The system will be used through a web browser. Students will see a login page, a list of scholarships for their program, a scholarship details page, an application form and an application status page. Staff will see pages to manage scholarships, review applications, schedule interviews and record approvals. All pages will have a consistent look, easy and concise wording and clear error messages for better usability.

2.1.3 Hardware Interfaces

The system will run on university server. Students and staff can use any electronic device that has a web browser and a network connection.

2.1.4 Software Interfaces

The server needs an operating system, a database (e.g. MySQL) and a web server. Users need a latest web browser such as Chrome, Edge or Firefox. The technologies are listed in Section 3.1.3.

2.1.5 Communications Interfaces

Users connect to the system through the university network or the internet using https, so that data is sent securely. Status changes are shown inside the system when the user logs in.

2.1.6 Memory Constraints

There are no special memory limits. The system mainly stores text records and uploaded documents (PDF or image files), so the server needs enough storage space. Each uploaded file will be limited to 2 MB.

2.1.7 Operations

Staff will add and publish scholarships, review applications and record approval status. Students will apply while a scholarship is open. The database will be backed up regularly and can be restored if data is lost.

2.1.8 Site Adaptation Requirements

Before the system is used, the university must provide the list of academic programs, student records, staff accounts and the eligibility rules of each scholarship.

2.2 Product Functions

Summary of the major functions the system will perform:

- Student Authentication: Authenticate students according to their academic program.
- Scholarship Categorization: Categorize scholarships into Need-Based and Merit-Based Scholarships according to programs.
- Eligibility Criteria: Define and check the eligibility criteria for each scholarship.
- Program Based Eligibility: Allow students to submit applications for scholarships specific to their academic program.
- Interview Scheduling: Schedule interviews for eligible scholarship applicants when required.
- Application Approval: Manage the approval or rejection of scholarship applications.
- Scholarship Restriction: Ensure that a student cannot receive another scholarship once their application has been approved.
- Advertise Scholarships: Advertise available scholarships to inform students about scholarship opportunities and their application deadlines.

2.3 User Characteristics

User Class	Description	Technical Expertise	Privileges
Student	Students will have valid university credentials for authentication. Students will be able to view and apply for eligible scholarships through the system.	Basic computer and web browsing skills.	Log in, view scholarships of own program, check eligibility, apply, track own application status.
Authorized Staff	University staff members who are permitted to manage scholarship information and applications.	Basic office computer skills and short training about the system.	Add, update, delete and advertise scholarships; review applications; schedule interviews; approve or reject; view approval records.

User Class	Description	Technical Expertise	Privileges
System Administrator	Person who looks after the server, database and user accounts.	Good technical knowledge of servers and databases.	Manage accounts, take backups, restore data, maintain the system.

2.4 Constraints

- The system must work within the available budget and resources of the university.
- Only authorized staff will be allowed to add or update scholarship information.
- Students' personal information, documents, and financial background must remain confidential.
- The system should verify submitted documents to make sure they are authentic.
- The system will depend on the available hardware and software infrastructure of the university.
- Only authorized staff should be able to view or update sensitive application and approval information.

2.5 Assumptions and Dependencies

- The university will have the required hardware and software infrastructure to support the system.
- The scholarship system will be linked with the university administration to receive and display newly announced scholarships.
- Students will have valid university credentials for authentication.
- Scholarship eligibility criteria will be clearly defined and provided by the university.
- Authorized scholarship staff will update scholarship information and application status.
- Scholarship approval records will be maintained and kept updated in the system.
- Each student can have only one approved scholarship at a time.

2.6 Apportioning of Requirements

Requirements will be delivered incrementally according to the following schedule:

Tasks / Phases	Time Period	Deliverables
Planning	Week 1	Project Plan
Overall System Analysis & Design	Week 2	System Design
Increment 1 – Authentication & Scholarship Categorization	Weeks 3–4	Tested Increment 1
Increment 2 – Eligibility & Program-Based Eligibility	Weeks 5–6	Tested Increment 2

Tasks / Phases	Time Period	Deliverables
Increment 3 – Advertisement & Student Application	Weeks 7–9	Tested Increment 3
Increment 4 – Interview, Approval & Scholarship Restriction	Weeks 10–11	Tested Increment 4
Final Integration & System Testing	Week 12	Integrated & Tested System
Documentation & Submission	Week 13	Final Project

Features that are not essential for the first version can be added in a later version.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

The system will have the following main screens:

- Login screen for students and staff.
- Student home page showing scholarships for the student's program.
- Scholarship details page with scholarship type, eligibility criteria and last date.
- Application form with uploaded documents.
- Application status page.
- Staff pages: manage scholarships, publish advertisements, review applications, schedule interviews, approve or reject, and view approved records.

All screens will use a simple and consistent layout, clear labels, and error messages that tell the user what went wrong to enhance usability.

3.1.2 Hardware Interfaces

Server: A university server with at least 4 GB RAM and enough disk space for the database and uploaded documents.

Client: Any PC, laptop or smartphone with a web browser.

3.1.3 Software Interfaces

Tools identified in the Project Plan:

- Programming: Python, Java, or PHP
- Database: MySQL or MongoDB.
- Frontend: HTML, CSS, JavaScript.
- Project Management: Trello or Jira.
- Project Libre: For creating Gantt chart to set deadlines and track overall project scheduling.
- Java: Programming language having strong OOP features and wide community support, suitable for secure and scalable development.

3.1.4 Communications Interfaces

The browser communicates with the server using https. The server communicates with the database and with the university administration system over the university network.

3.2 Functional Requirements

3.2.1 Import Scholarship Information

Primary Actor: Admin

Preconditions:

- Admin is logged into the system.
- Admin has permission to import scholarship information.
- The external portals such as HEC portal or PEEF portal is available and contains scholarship information.
- External portals are accessible by the system.

Inputs:

- Selected external portals.
- Scholarship information
- Scholarship updates

Outputs:

- Imported scholarship information
- Updated scholarship information
- Import confirmation message

Main Success Scenario:

1. Admin selects Import Scholarship Information.
2. System displays the available external scholarship portals.
3. Admin selects one of the available external portals.
4. System connects to the external portal.
5. System retrieves available scholarship information and updates from the selected portal.
6. System validates the received scholarship information.
7. System displays the scholarship information to the admin.
8. Admin confirms the import of the scholarship.

9. System stores new scholarship information and updates existing scholarship information where needed.
10. System records the import status.
11. System displays an Import Successful message.

Alternative flows:

Alt flow 1:

- At step 6, the system identifies scholarship information has no new updates and is already present in the system, so it skips duplicate information.

Exception Flows:

- The system cannot connect to the external portal and displays an error message that portal is unavailable.
- System encounters some technical issue while importing the data. Therefore, system does not complete the import operation.

Post Conditions:

- New scholarship information is successfully stored in the system.
- The imported information is available for scholarship management.

3.2.2 Define Eligibility Criteria

Primary Actor: Admin

Pre-Conditions:

- Admin is logged into the system.
- Admin is authorised to manage scholarship eligibility.
- Scholarship has already been added to the system.

Inputs:

- Scholarship
- Eligibility criteria
- Academic program

Outputs:

- Define eligibility criteria
- Updated eligibility criteria
- Confirmation message

Main Success Scenario:

1. Admin selects Define Eligibility Criteria.
2. System displays the available scholarships information.
3. Admin selects a scholarship.
4. System displays the existing eligibility criteria of the selected scholarship.
5. Admin enters or updates the eligibility criteria.
6. Admin specifies the required academic program and other required conditions.
7. Admin confirms the criteria.
8. System validates the entered criteria.
9. System saves the eligibility criteria.
10. System displays Eligibility Criteria Saved Successfully confirmation message.

Alternative Flows:

Alt flow 1:

- At step7, admin cancels the criteria before its confirmation. Therefore, system cancels the action and no changes are saved.

Exception Flows:

- System cannot access the database therefore an error message is displayed, and eligibility criteria remains unchanged.

Post Conditions:

- Eligibility criteria are successfully stored for the selected scholarship.

3.2.3 Managing Scholarships

Primary Actor: Admin

Pre-Conditions:

- Admin is logged into the system.
- Admin has permission to manage scholarships.
- Required scholarship information is available.

Inputs:

- Scholarship Name
- Scholarship description
- Scholarship details
- Scholarship category

Outputs:

- Added Scholarship information
- Updated scholarship information
- Scholarship program
- Scholarship category
- Confirmation message

Main Success Scenario:

1. Admin selects Manage Scholarship.
2. System displays the scholarship management options.
3. Admin selects Add or Update Scholarship.
4. System displays the scholarship information form.
5. Admin enters or updates the scholarship information.
6. System includes Set Scholarship Program to link the scholarship with its academic program.
7. System includes Set Scholarship Category to categorise the scholarship as Need-Based or Merit-Based.
8. Admin confirms the scholarship information.
9. System validates the entered scholarship information.
10. System displays a Scholarship Saved Successfully confirmation message.

Alternative Flows:

Alt flow 1:

- System detects missing or invalid scholarship information.
- System displays an error message.
- Admin rectifies the information.
- System validates the information again.

Exception flows:

- System undergoes a technical error while saving the scholarship therefore scholarship information is not saved and system displays an error message.

Post Conditions:

- Scholarship information is successfully stored or updated in the system.
- The scholarship is categorized as Merit-Based or Need-Based.

3.2.4 Shortlisting Applications

Primary Actor: Scholarship Committee Member

Pre-Conditions:

- Scholarship Committee member is logged into the system.
- Scholarship applications have been submitted by students.
- Applications are available for reviewing.

Inputs:

- Application Information
- Student information
- Eligibility information
- Shortlisting decision

Outputs:

- Shortlisted Applications
- Updated application status
- Confirmation message

Main Success Scenario:

1. Scholarship committee selects Shortlist Applications.
2. System displays the submitted scholarship applications.
3. Scholarship committee member selects an application to review.
4. System displays the student’s application and eligibility information.
5. Scholarship committee member reviews the application.
6. Scholarship committee member selects the application that meet all the requirements for shortlisting.

7. System updates the selected applications status to Shortlisted.
8. System displays a confirmation message.

Alternative Flows:

Alt flow 1:

- At step5, scholarship committee reviews an application but it does not meet the required criteria so it doesnot get shortlisted.

Exception Flows:

- System cannot retrieve the required application information.
- System displays an error message.
- Committee cannot the shortlist the application until information is available.

Post Conditions:

- Selected scholarships have **Shortlisted** status.
- Those applications are available for further scholarship process.

3.2.5 "View Advertisement"

Primary Actor: Student

Inputs:

- Student_Id

Outputs:

- Scholarship title
- Description
- Eligibility criteria
- Application deadline

Pre-Conditions:

- The student is an authenticated user of the system.
- Scholarship advertisements are available in the system.

Main Success Scenario:

1. Students open the scholarship advertisements section.
2. System displays the list of available scholarship advertisements.
3. Students select a scholarship advertisement.
4. System displays the scholarship details.
5. System displays the scholarship category and academic program.

6. System displays the eligibility criteria.
7. System displays the application deadline.
8. Students read advertisement and scholarship information.
9. System provides the option to apply for the scholarship.

Alternative Flows:

- **Alt Flow A:** At step 2, no scholarship advertisements are currently available. System displays a message that no scholarships are available.
- **Alt Flow B:** At step 9, the student is not eligible due to program requirements. System does not allow the student to apply.

Exception Flows:

- **Exception 1:** System is unable to load scholarship advertisements. System displays an error message.
- **Exception 2:** System is unable to retrieve the selected scholarship details. System displays an error message.

Post Conditions:

- The selected scholarship advertisement and its relevant information are displayed to the student.
- The student can proceed to apply if eligible and the application is open.

3.2.6 The Use Case: "View Application Status"

Primary Actor: Student

Inputs:

- Student_Id
- Application_Id

Outputs:

- Application details
- Scholarship details
- Status message

Pre-Conditions:

- The student is an authenticated user of the system.
- The student has submitted a scholarship application.

Main Success Scenario:

1. Students open the application status section.

2. System displays the student's submitted scholarship application.
3. Students select an application.
4. System retrieves the selected application details.
5. System displays the scholarship details.
6. System displays the current application status.
7. System displays any relevant status message or update.
8. Student view the application status.

Alternative Flows:

- **Alt Flow A:** At step 2, the student has submitted multiple applications. System displays all submitted applications.
- **Alt Flow B:** At step 2, the student has no submitted applications. System displays a message that no applications are available.
- **Alt Flow C:** At step 3, the selected application is no longer available. System displays a message that the application cannot be found.

Exception Flows:

- **Exception 1:** System is unable to retrieve the application information. System displays an error message.
- **Exception 2:** System encounters a technical issue while loading the application status. System displays an error message.

Post Conditions:

- The current status of the selected scholarship application is displayed to the student.
- No application information can be changed.

3.2.7 The Use Case: "Advertising Scholarships"

Primary Actor: Scholarship Staff Member

Inputs:

- Scholarship_Id
- Scholarship_Title
- Scholarship_Description
- Scholarship_Category
- Academic_Program
- Eligibility_Criteria
- Application_Deadline
- Advertisement_Data

Outputs:

- Advertisement_Id

- Advertisement details
- Confirmation Message
- Error Message

Pre-Conditions:

- The scholarship staff member is an authorized user of the system.
- The scholarship information is available and verified.
- The application deadline and eligibility criteria have been defined.

Main Success Scenario:

1. Scholarship staff member opens the scholarship advertisement section.
2. System displays the option to create a new scholarship advertisement.
3. Scholarship staff member enters the create a new advertisement option.
4. System checks the access level of the user.
5. System displays the advertisement information form.
6. Scholarship staff member enters the scholarship information.
7. Scholarship staff member enters the scholarship category and academic program.
8. Scholarship staff member enters the eligibility criteria.
9. Scholarship staff member enters the application deadline.
10. Scholarship staff member submits the advertisement.
11. System checks that all required information is complete and valid.
12. System assigns an Advertisement ID.
13. System publishes the scholarship advertisement.
14. System displays a confirmation message to the scholarship staff member.
15. Students can view the published scholarship advertisement.

Alternative Flows:

- **Alt Flow A:** At step 11, required information is incomplete. System asks the staff member to complete the missing information.
- **Alt Flow B:** At step 11, the scholarship information is invalid. System asks the staff member to correct the information.
- **Alt Flow C:** At step 10, the staff member saves the advertisement as a draft instead of publishing it. System saves the advertisement as a draft.

Exception Flows:

- **Exception 1:** Scholarship staff member is not authorized to advertise scholarships. System refuses access.
- **Exception 2:** System is unable to save or publish the advertisement due to a technical issue. System displays an error message.

Post Conditions:

- The scholarship advertisement is saved and published.
- An Advertisement ID is assigned.
- Students can view the published scholarship advertisement and its application deadline.

3.2.8 "Submitting Application"

Actor: Student

Pre-condition:

- The student is an authenticated user of the system and logins with correct credentials. The scholarship is advertised and is open for applications.

Inputs:

- Student_ID
- Scholarship_ID
- Application_Data
- Documents

Outputs:

- Application_ID
- Application_Status
- Submission_Confirmation

Main Success Scenario:

1. Student opens the application form of the preferred scholarship.
2. System displays the application form.
3. Student reads the eligibility criteria of the scholarship.
4. Student enters the required data in the application form.
5. System automatically saves the entered progress.
6. Student attaches the required documents.
7. Student clicks the “Submit Application” option.
8. System checks that all required fields and documents are complete and valid.
9. System checks that the deadline has not passed, the student is eligible by category and program, the student has no approved scholarship, and the same application has not already been submitted.
10. System saves the application and assigns an Application ID.
11. System sets the application status to Pending.
12. System displays the submission confirmation, including the Application ID, to the student.

Alternative Flows:

- *Alt Flow A:* Student is notified that he is not eligible for scholarship due to category or program.

- *Alt Flow B:* Required fields are incomplete
- *Alt Flow C:* Required documents are incomplete

Exception Flows:

- *Exception 1:* Student has already submitted the same application.
- *Exception 2:* Student has failed to successfully submit application because deadline has passed.

Post Condition:

The application is saved with status Pending, an Application ID is assigned, and the submitted documents are linked to the application.

3.2.9 "Approve Scholarships"

Primary Actor: Scholarship Committee Member

Inputs:

- Application_ID
- Approval_Choice
- Student_ID

Outputs:

- Student details
- Application details
- Approval status
- Confirmation Message
- Error message
- Approval email

Pre-Conditions:

- The committee member is logged into the system.
- The application has been reviewed and all required documents have been checked.
- The application status is either Pending or Shortlisted.

Main Success Scenario:

1. Committee member opens the list of shortlisted applications.
2. Committee member selects an application.
3. System displays the applicant's details and application details.
4. Committee member reviews the final information.
5. Committee member selects the Approve option.

6. System checks whether the student already has an approved scholarship.
7. System updates the application status to Approved.
8. System sends an approval email to the student's email address.
9. System shows a confirmation message to the committee member.

Alternative Flows:

- *Alt Flow A:* At step 5, Committee member decides to reject scholarship which continues in the Reject Scholarship use case.
- *Alt Flow B:* At step 5, Committee member needs an interview first therefore Schedule Interviews use case is started and approval waits for result.

Exception Flows:

- *Exception 1:* Student already has an approved scholarship. In this case, system will refuse the approval and status will remain unchanged.
- *Exception 2:* System is unable to sent approval email. System displays an error message to the committee member.
- *Exception 3:* System cannot complete the approval process due to some technical issue and displays an error message.

Post Conditions:

- Application status is approved.
- An approval email has been sent to the applicant.
- The applicant cannot receive more than one scholarship.

3.3 Performance Requirements

The system shall meet the following performance requirements:

- The system shall support at least 200 simultaneous users.
- 95% of login requests shall be completed within 3 seconds under normal network conditions.
- 95% of page requests shall be completed and displayed within 3 seconds under normal network conditions.
- 95% of eligibility-check requests shall be completed within 2 seconds.
- 95% of application-submission requests shall be processed within 5 seconds, excluding document upload time.
- The system shall be capable of storing records for at least 10,000 students.
- Under peak workload conditions, the system shall support at least 200 simultaneous users without exceeding the specified response-time limits.

3.4 Logical Database Requirements

The system will store the following main data:

Entity	Main attributes	Relationship
Student	Registration number, name, program, CGPA, family income, scholarship status	Submits many applications
Scholarship	ScholarshipID, name, type, program, amount, seats, last date, eligibility criteria	Receives many applications
Application	ApplicationID, student, scholarship, date, documents, status	Belongs to one student and one scholarship
Interview	StudentID, application, date, time, venue	Belongs to one application
Approved Scholarship	Student, scholarship, approval date	Only one record per student
Staff	StaffID, name, role, login details	Reviews and approves applications

- Each student registration number must be unique.
- A student cannot have more than one approved scholarship.
- Application and approval records shall be kept for at least 1 year.
- The ER diagram is given in Appendix B.

3.5 Design Constraints

3.5.1 Standards Compliance

- This SRS follows IEEE Std 830-1998.
- Project management follows ISO/IEC/IEEE 16326.
- The system shall follow the university's own scholarship rules and policies.

3.5.2 Other Design Constraints

- The system will be web-based and developed using the incremental process model.
- The system must work within the available budget and infrastructure of the university.
- Only authorized staff may add or change scholarship information.

3.6 Software System Attributes

3.6.1 Reliability

- Take regular backups of the database.
- Keep a copy of important student and application records.
- Test the backup and recovery process when required.

3.6.2 Availability

The system should be available at least 98% of the time, and always during the scholarship application period.

3.6.3 Security

- Provide login access only to authorized users.
- Use different access levels for students and scholarship staff.
- Protect sensitive student and scholarship information from unauthorized access.
- Passwords shall be stored in encrypted form.
- Users shall be logged out automatically after a period of inactivity.

3.6.4 Maintainability

The system will be built as separate modules (one for each increment), so a change in one part does not break the others. The code will be commented and documented.

3.6.5 Portability

The system shall work on common browsers (Chrome, Edge, Firefox) and on Windows, Linux and macOS computers. The database can be moved to another server if needed.

3.7 Other Requirements

3.7.1 Data Validation

- Use validation checks to reduce incomplete data.
- Allow authorized staff to update incorrect records.

3.7.2 Error Handling

- Display clear error messages when information is missing or incorrect.

3.7.3 Legal, Documentation and Other Requirements

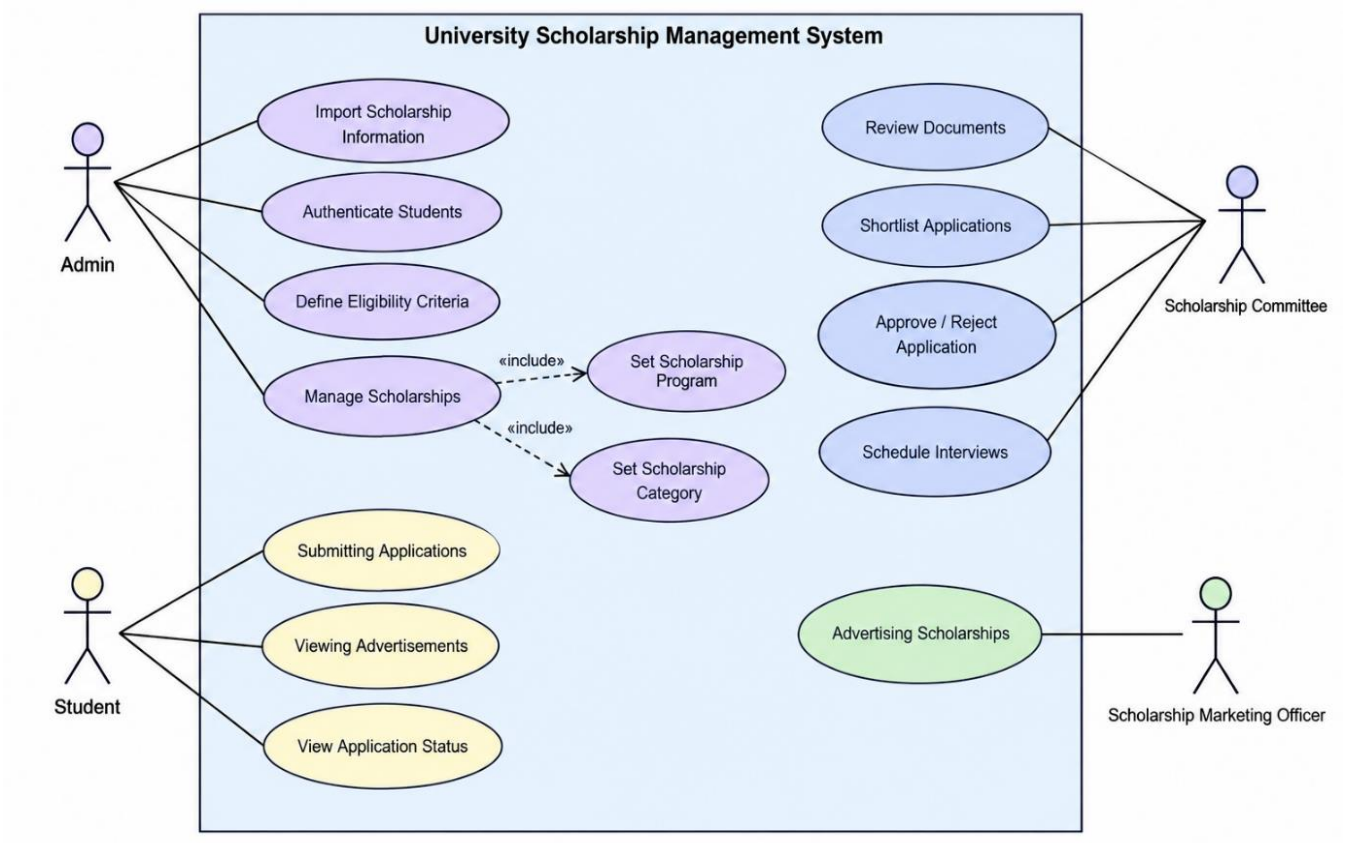
- Students' personal documents and financial information shall be kept confidential.
- The system will be tested with User Acceptance Testing (UAT) by scholarship staff and students.
- Final approval will be given by the university scholarship section after successful testing.

Appendices

Appendix A: Use Cases

Technique identified in the Project Plan: Use case diagrams for requirements.

Use Case	Actor
Login	Student, Staff
Manage Scholarships	Authorized Staff
Publish Scholarship Advertisement	Authorized Staff
View Scholarships	Student
Check Eligibility	Student, System
Submit Application	Student
Track Application Status	Student
Schedule Interview	Authorized Staff
Review Application	Authorized Staff
Approve or Reject Application	Authorized Staff, System



Appendix B: Analysis Models

